

Cardiovascular System

Part I: Multiple Choice Questions (MCQs)

Section 1: General Anatomy and Circulation

1. The cardiovascular system primarily consists of which two components?
 - A. Lungs and Heart
 - B. The heart and blood vessels
 - C. Arteries and Veins only
 - D. Atria and Ventricles only
2. Which side of the heart is responsible for pumping deoxygenated blood to the lungs?
 - A. The right side pump
 - B. The left side pump
 - C. Both sides pump deoxygenated blood
 - D. Neither side; the lungs pump the blood
3. The systemic circulation is powered by the:
 - A. Right atrium
 - B. Right ventricle
 - C. Left side pump
 - D. Pulmonary valve
4. Which chambers are known as the "receiving chambers" of the heart?
 - A. Ventricles
 - B. Atria
 - C. Aorta
 - D. Pulmonary arteries

Section 2: Valves and Heart Sounds

5. The Atrioventricular (AV) valves are located between the:

- A. Ventricles and Arteries
 - B. Atria and Ventricles
 - C. Left and Right atria
 - D. Pulmonary and Aortic arteries
6. The "Lub" sound of the heart (the first heart sound) is caused by the closure of:
- A. Semilunar valves
 - B. AV valves (Mitral and Tricuspid)
 - C. The Aorta
 - D. The S-A node
7. The second heart sound ("dub") occurs when the:
- A. Atria contract
 - B. Ventricles relax and semilunar valves close
 - C. Ventricles contract
 - D. Mitral valve closes

Section 3: Heart Wall and Contraction Mechanism

8. The muscular middle layer of the heart wall is called the:
- A. Epicardium
 - B. Endocardium
 - C. Myocardium
 - D. Pericardium
9. Which ion is released from the sarcoplasmic reticulum to promote the interaction of actin and myosin for contraction?
- A. Sodium (Na^+)
 - B. Calcium (Ca^{2+})
 - C. Potassium (K^+)
 - D. Magnesium (Mg^{2+})

Section 4: Electrical Activity and Properties

10. Which part of the conducting system is known as the "pacemaker"?

- A. AV node
- B. Sinoatrial (SA) node
- C. Purkinje fibers
- D. Bundle of His

11. "Automaticity" is defined as the heart's ability to:

- A. Respond to electrical stimulus
- B. Beat regularly
- C. Generate electrical stimulus
- D. Conduct impulses from one part to another

12. The ability of the heart to beat regularly (average 72 beats/min) is called:

- A. Excitability
- B. Rhythmicity
- C. Conductivity
- D. Contractility

Section 5: Cardiac Cycle and Loads

13. During Diastole, which of the following occurs?

- A. Ventricles contract and push blood to the body
- B. Blood is collected in the atria and pushed into resting ventricles
- C. The aortic valve opens
- D. Peripheral resistance is at its peak

14. The load that makes the ventricular muscles reach their maximum length at the end of diastole is:

- A. Preload
- B. Afterload

- C. Peripheral resistance
- D. Blood pressure

15. Afterload depends primarily on:

- A. Venous return
 - B. Resistance of the arteries (peripheral resistance)
 - C. The volume of blood in the atrium
 - D. The SA node signal strength
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Part II: Essay Questions

1. Describe the flow of blood through the two separate pumps of the heart, specifying the type of blood (oxygenated/deoxygenated) and the circulation involved for each side.
 2. Explain the mechanism behind the two heart sounds ("lub-dub") and identify which valves are associated with each sound.
 3. List and briefly define the five physiological properties of the heart.
 4. Compare and contrast "Preload" and "Afterload." Include what they depend on and when they occur in the cardiac cycle.
 5. Outline the pathway of the electrical signal in the heart, starting from the pacemaker to the ventricles.
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Answer Key

Part I: MCQs

1. B | 2. A | 3. C | 4. B | 5. B | 6. B | 7. B | 8. C | 9. B | 10. B | 11. C | 12. B | 13. B | 14. A | 15. B

Part II: Essay Model Hints

1. Right side: Deoxygenated blood to lungs (Pulmonary circulation). Left side: Oxygenated blood to tissues (Systemic circulation).

2. Lub (1st): Ventricles contract, AV valves (Mitral/Tricuspid) close. Dub (2nd): Ventricles relax, Semilunar valves (Aortic/Pulmonary) close.
3. Automaticity (Generate stimulus), Rhythmicity (Regular beating), Contractility (Push blood), Excitability (Respond to stimulus), Conductivity (Conduct stimulus).
4. Preload: Load at end of diastole; depends on venous return. Afterload: Resistance to open aortic valve; depends on peripheral resistance.
5. Pathway: SA node → Atria contract → AV node → AV Bundle (Bundle of His) → Bundle branches → Purkinje fibers → Ventricles contract.